# FitBuddy – Project Design Phase

**Project Name: FitBuddy - AI Fitness Plan Generator using Gemini Models**

## Detailed Project Documentation

## 1. Design Overview

The FitBuddy design divides the application into presentation, application logic, AI integration and optional data-storage layers.

### *Tabulation*

| Item | Details |
|---|---|
| Project | FitBuddy |
| AI Technology | Gemini Models |
| Type | Generative AI application |
| Purpose | Personalized wellness/activity planning |

### *Coding / Technical Example*

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

### *Picture / Diagram*

```
IDEA → USER NEED → AI SOLUTION → FITBUDDY
```

## 2. Architecture

Frontend → Backend → Prompt Builder → Gemini API → Response Processor → User Interface. An optional database can store profiles and generated plans.

### *Tabulation*

| Requirement | Description |
|---|---|
| Functional | Collect input, generate and display plan |
| AI | Gemini API integration |
| Security | Protect API credentials |
| Usability | Simple and clear interface |

### *Coding / Technical Example*

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
```

```
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

### *Picture / Diagram*

```
USER REQUIREMENTS → FUNCTIONAL + NON-FUNCTIONAL → SYSTEM REQUIREMENTS
```

## 3. User Interface Design

The home page can contain the FitBuddy title, a short description, input fields, a Generate Plan button and a result area. The result should be organized using headings, cards or tables.

### *Tabulation*

| Layer | Main Responsibility |
|---|---|
| Frontend | Collect and display information |
| Backend | Validation and application logic |
| AI Layer | Gemini prompt and generation |
| Database | Optional plan storage |

### *Coding / Technical Example*

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user_input = {
    "experience": "beginner",
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# Display the generated response in the UI.
```

### *Picture / Diagram*

```
FRONTEND → BACKEND → GEMINI API → RESPONSE → UI
```

## 4. Backend Design

The backend receives form data, validates it, builds a controlled prompt, calls Gemini, checks the returned content and sends the formatted result back to the frontend.

### *Tabulation*

| Feature | Purpose |
|---|---|
| Profile | Capture user preferences |
| Activity | Choose preferred activity |
| Time | Specify available duration |
| Generate | Request AI plan |

### *Coding / Technical Example*

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

### *Picture / Diagram*

```
PROFILE FORM → VALIDATION → GENERATE BUTTON → PLAN
```

## 5. Gemini Prompt Design

A good prompt should specify the role of the model, user information, desired plan format, general wellness focus, safety boundaries and output structure.

### *Tabulation*

| AI Element | Role |
|---|---|
| Prompt | Defines task and constraints |
| Gemini | Generates natural-language output |
| Response | Returned AI content |
| Formatter | Creates readable result |

### *Coding / Technical Example*

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
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}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

### *Picture / Diagram*

```
USER DATA → PROMPT → GEMINI → AI RESPONSE → FORMATTER
```